

Wi-Fi Sensing based Real-Time Activity Detection in Smart Home Environment

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Abstract—Wi-Fi sensing technology is being used extensively for different sensing applications, mostly for human activity recognition in the recent past. The Channel State Information (CSI) identifies the frequency shifts in the wireless medium caused by movements and changes in the region of interest, which can be analyzed using the amplitude data of various frequency channels. However, the existing systems are not suitable for real-time implementation due to the three layer cloud based architecture used. With IoT devices, edge computing provides a suitable solution with negligible communication latency and reduced network traffic. The proposed work mainly focused on real-time human activity information extraction in smart home environments using CSI values of low-cost ESP32 WiFi device. The center point of this work is to implement the IoT and edge layers of the three layered architecture to extract, process and visualize the sensing data in real-time with low-latency. We use simple statistical features and light weight machine learning algorithms for human activity recognition in real-time instead of complex and computationally heavy algorithms which is not suitable for edge computing.

Index Terms—WiFi sensing, channel state information, edge computing, activity recognition

I. INTRODUCTION

Human Activity Recognition (HAR) has many uses, including surveillance, patient monitoring in hospitals, fall detection for older adults, people counting, etc. Gaming, security, and enhancing human-computer interaction are all in scope [1]. While proximity, pressure, and light sensors gave acceptable results, their intrusive nature presented challenges. Due to the capacity of radio frequency based systems to take advantage of the current WiFi signals to sense the environmental dynamics.

WiFi channel is established between the transmitter and receiver and any movements in this region of interest will influence the signal propagation. This information captured by the signal subcarriers of CSI to analyze the change patterns. CSI is the physical layer information and can encapsulate the environmental attenuation, distance attenuation, signal scattering, and diffraction effects on the WiFi signals. The complex CSI data contain amplitude and phase variations across multiple subcarriers.

This work focuses to build a simple, low-cost activity recognition system that is easy to deploy in real-world situations. This can be accomplished with the help of fine-grain CSI data

with low-cost IoT devices without using high-level resources, proposing architecture for real-time data transfer, processing, visualizing, and identifying patterns in the collected CSI data.

The main contributions of this research are as follows:

- Proposed a two layer architecture for real-time activity detection
- Used simple statistical features to minimize the latency and deploy the model in low-cost resource constrained ESP32 modules

II. RELATED WORKS

A detailed survey of activity recognition research are mainly categorized into vision-based and sensor-based systems [2]. In vision-based sensing methods, computer vision techniques needed for processing, which adds more processing time and costly hardware. Performance of this sensing method may raise privacy concerns for the users and not efficient in low light conditions. Wearable sensors like accelerometer and pressure sensors are used in sensor-based methods to collect data, but are not comfortable and acceptable for users.

The state-of-the-art techniques for activity detection classified into two categories - coarse-grain and fine-grain approaches [1]. Received Signal Strength Information (RSSI) is used for identifying people based on their body shape in coarse-grain activity detection. Fine-grain activity detection systems mainly use CSI and can detect high-motion activities like squatting, walking, sitting etc [3]. Other applications include respiration and sleep monitoring.

Popular WiFi sensing tool Intel 5300 network interface controller (NIC) based on Linux 802.11n CSI tool is outdated now. Also. This tool is not suitable for IoT applications because of high deployment cost, cannot function as standalone devices, and are relatively vulnerable and prone to hardware failures. These devices also provide low-resolution CSI data with less subcarriers. To overcome these shortcomings, we presented a low-cost standalone device called ESP32 micro-controller [4], which overcome the above issues.

Several approaches to the activity recognition problem include model-based, learning-based, and pattern-based approaches [5]. Pattern-based recognition aims to find different patterns, extract compelling features and recognize human activities. The model-based approach recognizes the activities

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and behaviour using physical-mathematical models. However, these methods are challenging to implement where the whole environment cannot be depicted as a mathematical model. Deep learning-based recognition has advantages like automatic feature discovery but is time-consuming.

III. METHODOLOGY

Generally, IoT devices generate enormous amounts of data. These generated data are processed and analyzed by cloud services. However, sending this massive amount of data to the cloud increase latency is not cost-effective, and is not ideal for real-time applications. Edge computing enables basic computations close to the data sources, reducing the data transported to the cloud for final processing and analysis and making the system cost-efficient. It reduces latency, improves user experience, gives better and faster insights.

A. Proposed Architecture

In this paper, we propose a novel two layer architecture model for real-time activity detection using edge computing technology. Fig. 1a shows a general 3-layer IoT-edge-cloud architecture. The bottom layer is the IoT layer comprising physical devices that communicate, generate, send and receive data. It is also called the data source layer. The middle layer is the edge computing layer that collects data from the IoT layer and processes it. The devices in this layer filter, visualize and process the data almost in real-time. The top layer is the cloud layer, whose services include big-data analysis and data warehousing. It gets data from the edge layer.

We mainly concentrate on IoT and edge layers in this work and the low-cost ESP32 used both at the IoT and Edge layer. Fig. 1b is the flow diagram of the proposed IoT-edge architecture. The receiver in the IoT layer collects the CSI data and sends it to the ESP32 in the edge layer using UART ports available for ESP32 devices. The edge layer extracts amplitude and phase from the raw CSI data, pre-process the amplitude data by removing pilot and null sub-carriers, and filters used to remove outliers and noise. Then simple classifier model is used to classify and real-time visualize classification results. After first-level detection at the edge layer, then the extracted features are sent to the cloud for further computation and analysis.

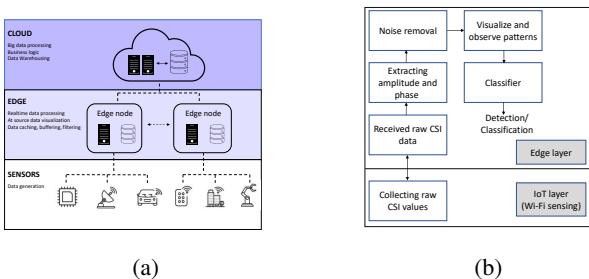


Fig. 1: (a) General 3-layer IoT architecture and (b) proposed architecture

IV. EXPERIMENTAL SETUP, DATA COLLECTION, ANALYSIS AND RESULTS

We used three low-cost ESP32 controllers, two of them used in IoT layer and one of device used in edge layer. The ESP-32 module at IoT layer estimates the channel to extracts the CSI values and communicates to the ESP32 module in the Edge layer for further analysis.

A. Data Collection

In IoT layer one of the ESP32 device acts as an Access Point (AP) or transmitter (Tx) and the other one act as a receiver (Rx) also called Station (STA). Experiments are conducted in a laboratory room (dimension 6m x 6m) and the distance between transmitter and receiver is 3 meters with 1 meter height from ground. The following activities are carried out in the WiFi region of interest - walking, jumping, falling and sitting. The CSI data continuously collected through the serial port at the receiver (STA) end.

B. Data transfer from IoT layer to Edge layer

We used the Universal Asynchronous Receiver/Transmitter (UART) hardware feature supported by the ESP32. The data is transferred in the form of bits from one device to another. The two ESP32 devices have to be cross-connected such that the T_x of one is bound to the R_x of the other and vice versa. The receiver ESP32 (STA) in the IoT layer collects data from the transmitter (AP) and simultaneously sends it to the ESP32 in the edge layer. It writes the data into the T_x buffer using `uart_write_bytes()` and waits for the complete buffer data to be transmitted. The T_x buffer is flushed between each frame transfer. The edge layer ESP32 is configured with UART parameters identical to that of the IoT layer STA. It reads the data of specified bytes using `uart_read_bytes()`. The R_x buffer is flushed each time after receiving each frame. No significant delay in the transmission is observed.

C. Processing and Filtering the CSI data

Each received frame has 128 values corresponding to the real and imaginary parts of the complex CSI data [4]. We extracted 64-amplitude and 64-phase values from the CSI data per received frames. We focus only on amplitude values as the ESP32 is a low-end IoT device with limited processing capability. The phase values we get from the CSI data do not add any extra information. Moreover, it becomes an overhead to the activity event. Out of 64-subcarrier values of amplitude data, 11 are zero padded subcarriers, and the DC (NULL) sub-carrier is situated at the centre of the OFDM system. Among the remaining 52, four pilot signals are situated at positions at -21, -7, +7 and +21. We are left with 48 data subcarriers, which contain actual information for further analysis. We tried many filters, but Hampel filter gives best result for removing outliers and noise [6]. Fig. 2 shows the data before and after filtering.

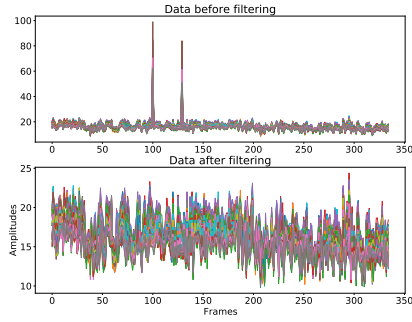


Fig. 2: Amplitude data before (top) and after filtering (bottom) for jumping activity

D. Data Analysis

We visualize the data to see how the values change across the subcarriers, frames and blocks (group of frames). We considered the following statistical parameters for analyzing the amplitude data; (1) average, (2) maximum, (3) minimum and (4) variance. We consider these four statistical features as our objective is to detect the activity in real-time and also at the edge layer only without using any cloud infrastructure.

1) *Subcarrier-wise analysis of CSI amplitude values*: The amplitude values are plotted for all 48-data subcarriers by calculating the average, maximum, or minimum frame values for that particular subcarrier. Fig. 3a shows subcarrier-wise analysis for same activity and Fig. 3b represents subcarrier-wise analysis for different activities.

2) *Frame-wise analysis of the CSI Amplitude of subcarriers*: The amplitude part of CSI values are plotted for each frame using the average, maximum, or minimum of all subcarriers values for that specific frame. Fig. 4a depicts the frame-wise analysis for same activity and Fig. 4b shows frame-wise analysis for different activities.

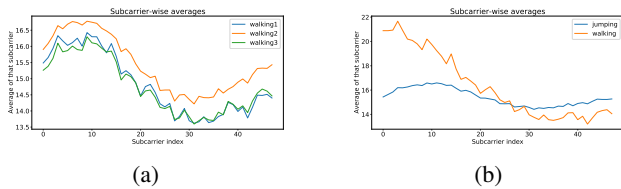


Fig. 3: Subcarrier-wise analysis for (a) walking and (b) waking and jumping activities

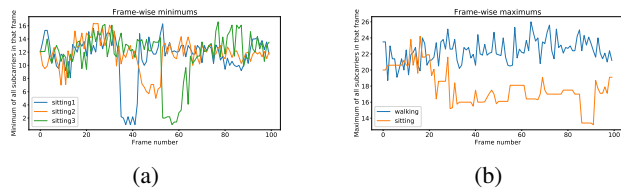


Fig. 4: Frame-wise analysis for (a) falling and (b) waking and sitting activities

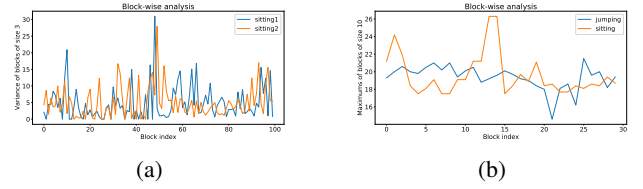


Fig. 5: Block-wise analysis for (a) sitting and (b) sitting and jumping activities

3) *Block-wise analysis of CSI amplitude values of subcarriers*: The entire data is segmented into blocks of a specific size (5 frames per block) to observe the variations across those blocks of frames. Average, maximum, minimum and variance metrics are considered for this analysis. Each metric is calculated for each block and plotted for the same and different activities. Fig. 5a shows block-wise analysis for same activity and Fig. 5b depicts block-wise analysis for different activities. All these graphs show that the CSI data exhibit similar patterns for the same activities and different patterns for different activities corresponding to all considered features.

E. Activity Classification using Machine Learning

In addition to visualization, the filtered data is used as input to a light-weight Support Vector Machine (SVM) machine learning classifier to classify activities at the edge layer. It classifies the test data into jumping, walking, sitting, and falling classes. The model showed an accuracy of around 70%. The prediction is then communicated to a mobile application for end user for real-time monitoring the activities.

V. CONCLUSIONS AND FUTURE WORK

In this work, we proposed and implemented Wi-Fi sensing system at bottom two-layers of the three layer IoT architecture that enables a lightweight, low-cost, real-time with low-latency sensing information of human activity in smart home environments. To make the entire system cost-effective and for ease of deployment, we used ESP32 devices both at IoT and edge layers. The experiments showed similar patterns for the same activity and different for different activities. We are working on development of smart phone based applications for better user-interface.

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